Notes: Bloom, Schankerman, and Van Reenen (Econometrica, 2013)

Identifying Technology Spillovers and Product Market Rivalry

Contents

1 Big picture

- **Question.** How large are R&D spillovers, and how can we separate positive technology spillovers from negative product-market rivalry? The paper studies the fact that another firm's R&D can help a firm through knowledge diffusion but hurt it through business stealing.
- **Core distinction.** The paper separates two kinds of "spillovers": *technology spillovers*, which raise other firms' knowledge, patenting, productivity, and value; and *product-market rivalry*, in which rivals' R&D reduces a firm's market value by threatening market share or margins.
- **Main empirical idea.** The authors locate firms in two distinct spaces. Technology space is based on patenting across technology classes. Product-market space is based on sales across four-digit industries. This produces two different spillover pools: *SPILLTECH* and *SPILLSIC*.
- **Identification logic.** The theory predicts different signs across outcomes. Technology spillovers should raise market value, patents, and productivity. Product-market rivalry should lower market value but need not raise patents or productivity. R&D responses are more ambiguous and reveal strategic complementarity or substitution.
- **Data setting.** The main data are a panel of publicly traded U.S. firms from Compustat matched to NBER patent data, covering firms with patenting histories and observations between 1981 and 2001.
- **Main result.** Technology spillovers and product-market rivalry both matter. Technology spillovers are quantitatively larger, so gross social returns to R&D exceed private returns.
- **Policy result.** In the baseline calculation, the private return to R&D is about 20.7%, while the social return is about 55.0%. This implies a large wedge and under-investment in R&D from the social point of view.
- **Small-firm result.** Smaller firms generate lower social returns in the sample because they tend to operate in narrower technological niches and are less connected to other firms in technology space.
- **Economic takeaway.** R&D policy should not treat all outside R&D as a single positive spillover pool. A firm's neighbors in technology space and its rivals in product space have different, offsetting effects.

2 Incremental contribution of the paper

- **Separating two effects previously mixed together.** Much of the R&D spillover literature measures outside R&D at the industry level. That mixes knowledge spillovers with rivalry. This paper separates the two by building different firm-to-firm proximity measures in technology space and product-market space.
- **A unified framework for multiple outcomes.** The paper derives sign predictions for market value, patents, productivity, and own R&D. This is important because no single outcome can cleanly identify the two mechanisms by itself.
- **Product-market distance at the firm level.** Building on Jaffe-style technology distance, the paper constructs an analogous product-market distance measure using line-of-business sales data. This creates within-industry variation that industry fixed effects alone cannot provide.

- **Mahalanobis distance contribution.** The authors develop a Mahalanobis proximity measure that allows knowledge to flow across related technology fields, not only within the same patent class. This is a methodological contribution beyond the baseline Jaffe measure.
- **Causal spillover strategy.** The paper addresses the reflection problem by instrumenting firm R&D with federal and state R&D tax incentives, then aggregating predicted R&D into technology and product-market spillover pools.
- **Private-versus-social return calculation.** The paper converts estimated coefficients into private and social marginal returns to R&D, incorporating both positive knowledge spillovers and negative business stealing.
- **Axiomatic comparison of distance measures.** The paper evaluates proximity measures using desirable properties such as economic microfoundations, scale invariance, within-field overlap, between-field overlap, and robustness to aggregation.
- **Policy contribution.** The analysis shows that larger and more technologically connected firms may generate higher social returns to R&D than small niche firms, which complicates the usual policy presumption that small firms should receive the most generous R&D subsidies on spillover grounds.

3 Data and measurement

3.1 Firm, patent, and accounting data

- The sample matches U.S. Compustat accounting and market-value data to NBER patent data.
- The panel covers 715 firms with at least four observations between 1980 and 2001 and at least one patent since 1963.
- Key firm variables include market value, fixed capital, sales, employment, R&D flows, and constructed R&D capital stocks.
- The R&D stock is built with the perpetual-inventory method:

$$G_t = R_t + (1-\delta)G_{t-1}, \qquad \delta = 0.15.$$

- Innovation output is measured with patent counts, especially citation-weighted patents, because patent values are highly heterogeneous.

Interpretation: The data are designed to connect firm performance, innovation, and market valuation with the external R&D performed by nearby firms in two different spaces.

3.2 Technology proximity and *SPILLTECH*

Let $T_i = (T_{i1}, \ldots, T_{i426})$ denote firm i's distribution of patents across technology classes. The baseline Jaffe technology proximity between firms i and j is

$$TECH_{ij} = \frac{T_i T_j'}{(T_i T_i')^{1/2}(T_j T_j')^{1/2}}.$$

The technology spillover pool for firm i is

$$SPILLTECH_{it} = \sum_{j \neq i} TECH_{ij} G_{jt},$$

where G_{jt} is firm j's R&D stock.

Interpretation: *SPILLTECH* is high when other firms with large R&D stocks are technologically close to firm i. It captures potential knowledge flows, not product competition.

3.3 Product-market proximity and *SPILLSIC*

Let $S_i = (S_{i1}, \ldots, S_{i597})$ denote firm i's sales distribution across four-digit industries. Product-market proximity is

$$SIC_{ij} = \frac{S_i S_j'}{(S_i S_i')^{1/2} (S_j S_j')^{1/2}}.$$

The product-market R&D exposure of firm i is

$$SPILLSIC_{it} = \sum_{j \neq i} SIC_{ij} G_{jt}.$$

Interpretation: *SPILLSIC* is high when firm i's product-market rivals perform large amounts of R&D. It is the measure that should capture business stealing and strategic rivalry.

3.4 Mahalanobis distance

The baseline Jaffe measure treats fields as orthogonal: firms are close only when they patent in the same class. The Mahalanobis extension allows related fields to matter. In simplified form,

$$TECH^M_{ij} = \frac{T_i \Omega T_j'}{(T_i T_i')^{1/2} (T_j T_j')^{1/2}},$$

where Ω summarizes how technology classes are co-located within firms.

Interpretation: The Mahalanobis approach recognizes that R&D in related technology areas can spill over even if patent classes are not identical.

3.5 Variation in the two distance measures

- The raw correlation between the pairwise product-market and technology proximity measures, SIC and $TECH$, is positive but far from one.
- The correlation between $\ln(SPILLTECH)$ and $\ln(SPILLSIC)$ is also positive but imperfect, and the within-firm changes are even less correlated.
- This imperfect correlation is essential: if technology closeness and product-market closeness were the same object, the paper could not separately estimate knowledge spillovers and rivalry.

Interpretation: The identifying variation comes from firms such as technology neighbors that are not product rivals, and product rivals that are not especially close technologically.

4 Models and empirical specifications

4.1 Analytical framework

The model has two stages. In stage 1, firms choose R&D, which produces knowledge. In stage 2, firms compete in the product market conditional on knowledge.

For firm 0, knowledge is produced as

$$k_0 = \phi(r_0, r_\tau),$$

where r_0 is own R&D and r_τ is R&D by a technologically close firm. Product-market rival R&D, r_m, affects profits through competition rather than through direct knowledge production.

The firm's value-maximization problem can be written as

$$\max_{r_0} V^0 = \Pi\big(\phi(r_0, r_\tau), k_m\big) - r_0.$$

Interpretation: The model separates the channel through which a technology neighbor helps the firm from the channel through which a product-market rival hurts the firm.

4.2 Core comparative statics

- Technology spillovers increase knowledge, patents, productivity, and market value.
- Product-market rivalry reduces market value because rivals' R&D creates business stealing.
- Product-market rivals' R&D should not directly increase the firm's patents or productivity, conditional on own R&D.
- The effect of other firms' R&D on own R&D is ambiguous. It depends on whether R&D choices are strategic complements or substitutes and on how technology spillovers affect the marginal return to own R&D.

Interpretation: The paper's identification is not based on one regression sign. It is based on a pattern of signs across several outcomes.

4.3 Market value equation

The market value equation augments the Griliches-style Tobin's Q framework with the two spillover pools:

$$\ln\left(\frac{V}{A}\right)_{it} = \phi\left(\frac{G}{A}_{it-1}\right) + \gamma_2 \ln SPILLTECH_{it-1} + \gamma_3 \ln SPILLSIC_{it-1} + X_{it}'\gamma + \eta_i + \tau_t + u_{it}.$$

Expected signs: $\gamma_2 > 0$ and $\gamma_3 < 0$. Technology neighbors raise value; product-market rivals reduce value.

4.4 Patent equation

The paper estimates citation-weighted patents using a count model:

$$P_{it} = \exp\{\lambda_1 \ln G_{it-1} + \lambda_2 \ln SPILLTECH_{it-1} + \lambda_3 \ln SPILLSIC_{it-1} + X_{it}'\lambda + \eta_i + \tau_t + u_{it}\}.$$

Expected signs: $\lambda_2 > 0$ and $\lambda_3 \approx 0$. Knowledge spillovers should raise innovation output; product-market rivalry should not mechanically raise patents.

4.5 Productivity equation

The production-function specification is

$$\ln Y_{it} = \varphi_1 \ln G_{it-1} + \varphi_2 \ln SPILLTECH_{it-1} + \varphi_3 \ln SPILLSIC_{it-1} + X_{it}'\varphi + \eta_i + \tau_t + u_{it},$$

where X includes production inputs such as labor and capital.

Expected signs: $\varphi_2 > 0$ and $\varphi_3 \approx 0$, although measured revenue productivity can be affected by firm-specific prices.

4.6 R&D equation

The R&D intensity equation is

$$\ln\left(\frac{R}{Y}\right)_{it} = \alpha_2 \ln SPILLTECH_{it-1} + \alpha_3 \ln SPILLSIC_{it-1} + X_{it}'\alpha + \eta_i + \tau_t + u_{it}.$$

Interpretation: The R&D equation is mainly about strategic responses. A positive coefficient on *SPILLSIC* suggests that product-market rivals' R&D and own R&D are strategic complements.

4.7 Private and social returns

The private marginal return includes the firm's own R&D productivity effect and the business-stealing effect of product-market rivalry. The social marginal return includes own R&D productivity and the technology spillovers generated for other firms. The key economic comparison is therefore

$$MSR - MPR \quad \text{depends on} \quad \text{technology spillovers} - \text{rivalry/business stealing effects}.$$

Interpretation: Product-market rivalry raises private incentives to do R&D by redistributing profits between firms, but it is not a social gain. Technology spillovers are social gains not fully captured by the innovating firm.

5 Identification and empirical design

5.1 What the design tries to identify

- The paper seeks to identify the separate effects of technology spillovers and product-market rivalry on firm value, innovation, productivity, and R&D.

- The key threat is the reflection problem: firms close in technology space may all increase R&D because of common technological opportunities, not because of spillovers.
- Firm fixed effects absorb time-invariant firm heterogeneity. Year effects absorb aggregate shocks.
- Industry demand controls help ensure that *SPILLSIC* is not merely capturing industry booms.

5.2 Instrumental-variable strategy

- The paper uses changes in federal and state R&D tax incentives to instrument for R&D.
- Firm-specific R&D tax prices vary because of federal rules and because states differ in their R&D tax treatment.
- The authors aggregate predicted R&D across technology and product-market proximity weights to instrument for *SPILLTECH* and *SPILLSIC*.
- The empirical logic is that tax-policy changes shift other firms' R&D supply, rather than directly shifting firm i's productivity or market value except through spillovers.

Interpretation: The tax-price instruments address the concern that spillover pools are correlated with unobserved opportunities or demand shocks. The exclusion restriction is still a strong assumption, but the IV results broadly reinforce the main sign pattern.

5.3 What the design cannot fully eliminate

- The proximity measures themselves are built from firms' technology and product-market choices, which may be endogenous over long horizons.
- Publicly traded Compustat firms are not representative of all firms, especially small private firms.
- The production-function estimates use real sales, so firm-specific price variation can contaminate productivity.
- The R&D equation is the least clean empirically because strategic complementarity may be confounded by common shocks.

Interpretation: The paper is strongest on the cross-outcome pattern: technology closeness predicts higher value, patents, and productivity, while product-market closeness predicts lower value and not higher patents or productivity.

6 Tables and figures

6.1 Table I and Figure 1: Theory and identifying variation

Read - Table I:

- Table I lays out the theoretical sign predictions for market value, patents/productivity, and R&D under different assumptions.

- Without technology spillovers or product-market rivalry, other firms' R&D should not affect the firm's value, knowledge, or R&D.
- With technology spillovers, *SPILLTECH* should raise market value and patents/productivity.
- With product-market rivalry, *SPILLSIC* should lower market value, but not directly affect patents/productivity.
- R&D responses are ambiguous because they depend on strategic complementarity/substitutability and on marginal returns in knowledge production.

Read - Figure 1:

- Figure 1 plots pairwise product-market closeness (SIC) against technology closeness ($TECH$).
- The scatter is dispersed across the unit square, not concentrated on a single diagonal.
- This means firms can be close in technology but far in product markets, or close in product markets but far in technology.

Economic message: Identification comes from separating technology neighbors from product rivals. The paper's central empirical move is to avoid treating all external R&D as the same type of spillover.

TABLE I

THEORETICAL PREDICTIONS FOR MARKET VALUE, PATENTS, AND R&D UNDER DIFFERENT ASSUMPTIONS

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
			No Technology Spillovers			Technology Spillovers		
Equation	Comparative Static Prediction	Empirical Counterpart	No Product Market Rivalry	Strategic Complements	Strategic Substitutes	No Product Market Rivalry	Strategic Complements	Strategic Substitutes
Market value	$\partial V_0/\partial r_\tau$	Market value with *SPILLTECH*	Zero	Zero	Zero	Positive	Positive	Positive
Market value	$\partial V_0/\partial r_m$	Market value with *SPILLSIC*	Zero	Negative	Negative	Zero	Negative	Negative
Patents (or productivity)	$\partial k_0/\partial r_\tau$	Patents with *SPILLTECH*	Zero	Zero	Zero	Positive	Positive	Positive
Patents (or productivity)	$\partial k_0/\partial r_m$	Patents with *SPILLSIC*	Zero	Zero	Zero	Zero	Zero	Zero
R&D	$\partial r_0/\partial r_\tau$	R&D with *SPILLTECH*	Zero	Zero	Zero	Ambiguous	Ambiguous	Ambiguous
R&D	$\partial r_0/\partial \mathrm{r}_m$	R&D with *SPILLSIC*	Zero	Positive	Negative	Zero	Positive	Negative

Notes: See text for full derivation of these comparative static predictions. Note that the empirical predictions for the (total factor) productivity equation are identical to the patents equation. Also note that the no technology spillovers case corresponds to $\phi_2 = 0$, and technology spillovers correspond to $\phi_2 > 0$. Strategic complementarity or substitutability between rivals' knowledge stocks is given by the sign of Π_{12}. *SPILLTECH* is the technology-distance weighted sum of all other firms R&D stocks. *SPILLSIC* is the product market-distance weighted sum of all other firms R&D stocks. See text for details.

Table I: theoretical predictions

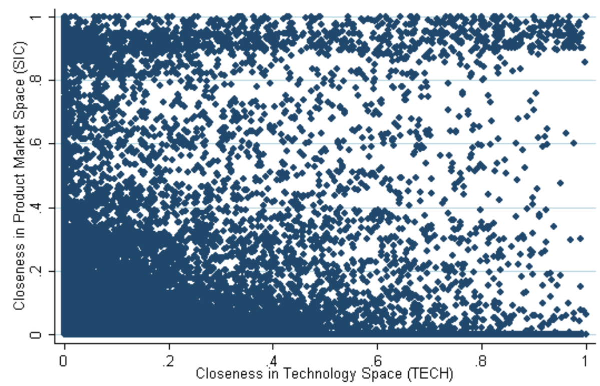

Figure 1: SIC and TECH correlations

6.2 Tables II and III: Sample scale and market-value effects

Read - Table II:

- Table II reports descriptive statistics. The median firm is large, with 3,839 employees, but the mean is much larger at 18,379, showing substantial skewness.
- The median R&D stock is \$28.7 million, while the mean is \$605 million, again showing the concentration of R&D among large firms.
- The median *SPILLTECH* is 20,091 and median *SPILLSIC* is 2,007, showing that the two exposure measures differ substantially in scale and dispersion.

Read - Table III:

- Without firm fixed effects, the signs are counterintuitive: *SPILLTECH* is negative and *SPILLSIC* is positive.
- With firm fixed effects, the signs align with theory: *SPILLTECH* is positive and *SPILLSIC* is negative.
- In the preferred fixed-effect Jaffe specification, a 10% increase in *SPILLTECH* is associated with about a 3.8% increase in market value, while a 10% increase in *SPILLSIC* is associated with about a 0.8% decrease.
- Mahalanobis and IV specifications produce larger absolute coefficients, consistent with attenuation in the simpler measure.

Economic message: Technology neighbors increase firm value; product-market rivals reduce it. This is the first central evidence that the two spillover channels point in opposite directions.

TABLE II
DESCRIPTIVE STATISTICS

Variable	Mnemonic	Median	Mean	Standard Deviation
Tobin's Q	*V*/*A*	1.41	2.36	2.99
Market value	*V*	412	3,913	16,517
R&D stock	*G*	28.7	605	2,722
R&D stock/fixed capital	*G*/*A*	0.17	0.47	0.91
R&D flow	*R*	4.36	104	469
Technological spillovers	*SPILLTECH*	20,091	25,312	19,942
Product market rivalry	*SPILLSIC*	2,007	6,494	10,114
Patent flow		1	16.2	75
Cite-weighted patents	*P*	4	116	555
Sales	*Y*	456	2,879	8,790
R&D-weighted sales/R&D stock	*Y*/*G*	2.48	3.83	19.475
Fixed capital	*A*	122	1,346	4,720
Employment	*N*	3,839	18,379	52,826

Notes: The means, medians, and standard deviations are taken over all non-missing observations between 1981 and 2001; values measured in 1996 prices in $million.

Table II: descriptive statistics

TABLE III
COEFFICIENT ESTIMATES FOR TOBIN'S Q EQUATION

	(1)	(2)	(3)	(4)	(5)	(6)
Specification:	OLS	OLS	OLS	OLS	OLS	IV 2nd Stage
Distance Measure:	Jaffe	Jaffe	Jaffe	Jaffe	Mahalanobis	Jaffe
$\ln(SPILLTECH_{t-1})$	−0.064	0.381	0.305		0.903	1.079
	(0.013)	(0.113)	(0.109)		(0.146)	(0.192)
$\ln(SPILLSIC_{t-1})$	0.053	−0.083		−0.050	−0.136	−0.235
	(0.007)	(0.032)		(0.031)	(0.050)	(0.109)
$\ln(\text{R\&D Stock/Capital Stock})_{t-1}$	0.859	0.806	0.799	0.799	0.835	0.831
	(0.154)	(0.197)	(0.198)	(0.198)	(0.198)	(0.197)
						1st Stage *F*-Tests
$\ln(SPILLTECH_{t-1})$						112.5
$\ln(SPILLSIC_{t-1})$						42.8
Firm fixed effects	No	Yes	Yes	Yes	Yes	Yes
No. observations	9,944	9,944	9,944	9,944	9,944	9,944

Notes: Dependent variable is ln(Tobin's Q) = $\ln(V/A)$ is defined as the market value of equity plus debt, divided by the stock of fixed capital. A sixth-order polynomial in $\ln(\text{R\&D Stock/Capital Stock})_{t-1}$ is included, but only the first term is shown for brevity. Standard errors in brackets are robust to arbitrary heteroskedasticity and first-order serial correlation using the Newey–West correction. A dummy variable is included for observations where lagged R&D stock is zero. All columns include a full set of year dummies and controls for current and lagged industry sales in each firm's output industry. Column (6) uses instrumental variable estimation. "1st Stage *F*-Tests" are the joint significance of the excluded tax-based instrumental variables ($\ln(TECHTAX)$ and $\ln(SICTAX)$) from each first stage of the endogenous variables, $\ln(SPILLTECH)$ and $\ln(SPILLSIC)$. See Appendix B.3 for details. In column (6), we also control for the firm's own federal and state R&D tax credit values.

Table III: Tobin's Q equation

6.3 Tables IV and V: Innovation and productivity effects

Read - Table IV:

- Citation-weighted patents are strongly related to own R&D stock.
- *SPILLTECH* is positive and significant across specifications, consistent with knowledge spillovers increasing innovation output.
- *SPILLSIC* is small and statistically insignificant, consistent with product-market rivalry not directly raising patent production.

Read - Table V:

- The productivity equation uses real sales as the dependent variable and controls for capital, labor, own R&D, and fixed effects.
- With firm fixed effects, *SPILLTECH* is positive and significant.

- *SPILLSIC* becomes insignificant once fixed effects are included, matching the theory prediction that product rivalry should not raise productivity.
- The negative *SPILLSIC* coefficient without fixed effects may reflect price or demand effects rather than true productivity.

Economic message: The innovation and productivity evidence supports the interpretation that *SPILLTECH* captures genuine knowledge flows, not merely correlated demand or rivalry.

TABLE IV

COEFFICIENT ESTIMATES FOR THE CITE-WEIGHTED PATENT EQUATION

	(1)	(2)	(3)	(4)	(5)
Specification:	Neg. Bin.	Neg. Bin.	Neg. Bin.	Neg. Bin.	Neg. Bin. IV 2nd Stage
Distance Measure:	Jaffe	Jaffe	Jaffe	Mahalanobis	Jaffe
$\ln(SPILLTECH)_{t-1}$	0.518	0.468	0.417	0.530	0.407
	(0.096)	(0.080)	(0.056)	(0.070)	(0.059)
$\ln(SPILLSIC)_{t-1}$	0.045	0.056	0.043	0.053	0.037
	(0.042)	(0.037)	(0.026)	(0.037)	(0.028)
$\ln(\text{R\&D Stock})_{t-1}$	0.500	0.222	0.104	0.112	0.071
	(0.048)	(0.053)	(0.039)	(0.039)	(0.020)
$\ln(\text{Patents})_{t-1}$			0.420	0.425	0.423
			(0.020)	(0.020)	(0.020)
Pre-sample fixed effect		0.538	0.292	0.276	0.301
		(0.046)	(0.033)	(0.033)	(0.032)
					IV 1st Stage F-Tests
$\ln(SPILLTECH)_{t-1}$					74.6
$\ln(SPILLSIC)_{t-1}$					15.0
Firm fixed effects	No	Yes	Yes	Yes	Yes
No. observations	9,023	9,023	9,023	9,023	9,023

Notes: Dependent variable is Cite-weighted patents. Estimation is conducted using the Negative Binomial model. Standard errors (in brackets) allow for serial correlation through clustering by firm. A full set of time dummies, four digit industry dummies, and lagged firm sales are included in all columns. A dummy variable is included for observations where lagged R&D stock equals zero (all columns) or where lagged patent stock equals zero (column (3)). Columns (2) to (5) include the "pre-sample mean scaling approach" to estimate fixed effects of Blundell, Griffith, and Van Reenen (1999). The Negative Binomial IV specification in column (5) implements a control function approach which includes the first five terms of the expansion of the residual for the first stage regressions. "1st Stage F-Tests" are the joint significance of the excluded tax-based instrumental variables (ln(*TECHTAX*) and ln(*SICTAX*)) from each first stage of the endogenous variables, ln(*SPILLTECH*) and ln(*SPILLSIC*). See Appendix B.3 for details.

Table IV: cite-weighted patent equation

TABLE V

COEFFICIENT ESTIMATES FOR THE PRODUCTION FUNCTION

	(1)	(2)	(3)	(4)	(5)
Specification:	OLS	OLS	OLS	OLS	IV 2nd Stage
Distance Measure:	Jaffe	Jaffe	Jaffe	Mahalanobis	Jaffe
$\ln(SPILLTECH)_{t-1}$	−0.022	0.191	0.186	0.264	0.206
	(0.009)	(0.046)	(0.045)	(0.064)	(0.081)
$\ln(SPILLSIC)_{t-1}$	−0.016	−0.005		−0.007	0.030
	(0.004)	(0.011)		(0.021)	(0.054)
$\ln(\text{Capital})_{t-1}$	0.288	0.154	0.153	0.156	0.152
	(0.009)	(0.012)	(0.012)	(0.012)	(0.012)
$\ln(\text{Labor})_{t-1}$	0.644	0.636	0.636	0.637	0.639
	(0.012)	(0.015)	(0.015)	(0.015)	(0.016)
$\ln(\text{R\&D Stock})_{t-1}$	0.061	0.043	0.042	0.043	0.041
	(0.005)	(0.007)	(0.007)	(0.007)	(0.007)
					1st Stage F-Statistic
$\ln(SPILLTECH)_{t-1}$					112.4
$\ln(SPILLSIC)_{t-1}$					51.2
Firm fixed effects	No	Yes	Yes	Yes	Yes
No. observations	9,935	9,935	9,935	9,935	9,935

Notes: Dependent variable is ln(Sales). Standard errors (in brackets) are robust to arbitrary heteroskedasticity and allow for first-order serial correlation using the Newey–West procedure. Industry price deflators are included and a dummy variable for observations where lagged R&D equals to zero. All columns include a full set of year dummies and controls for current and lagged industry sales in each firm's output industry. Column (5) uses instrumental variable estimation. "1st Stage F-Statistic" are the joint significance of the excluded tax-based instrumental variables (ln(*TECHTAX*) and ln(*SICTAX*)) from each first stage of the endogenous variables, ln(*SPILLTECH*) and ln(*SPILLSIC*). See Appendix B.3 for details.

Table V: production-function equation

6.4 Tables VI and VII: R&D responses and summary pattern

Read - Table VI:

- The R&D equation is less stable than the value, patent, and productivity equations.
- *SPILLSIC* is positive in several OLS specifications, suggesting that own R&D and product-market rivals' R&D may be strategic complements.
- Once spillovers are treated as endogenous using the tax-price instruments, the *SPILLSIC* coefficient becomes insignificant.
- *SPILLTECH* has no robust sign in the R&D equation, which is consistent with the theory's ambiguity.

Read - Table VII:

- Table VII summarizes the signs from the basic empirical specifications against the theoretical predictions.
- The main six predictions are supported: *SPILLTECH* is positive for market value, patents, and productivity; *SPILLSIC* is negative for market value and near zero for patents and productivity.

- Mahalanobis and IV results preserve the same qualitative pattern.

Economic message: The paper's core contribution is the coherent sign pattern across outcomes. R&D by technology neighbors looks like knowledge spillovers; R&D by product rivals looks like business stealing.

TABLE VI

COEFFICIENT ESTIMATES FOR THE R&D EQUATION

	(1)	(2)	(3)	(4)	(5)
Specification:	OLS	OLS	OLS	OLS	IV 2nd Stage
Distance Measure:	Jaffe	Jaffe	Jaffe	Mahalanobis	Jaffe
$\ln(SPILLTECH)_{t-1}$	0.079	0.100	−0.049	−0.176	0.138
	(0.018)	(0.076)	(0.042)	(0.101)	(0.122)
$\ln(SPILLSIC)_{t-1}$	0.374	0.083	0.034	0.224	−0.022
	(0.013)	(0.034)	(0.019)	(0.048)	(0.071)
$\ln(\text{R\&D/Sales})_{t-1}$			0.681		
			(0.015)		
					IV 1st stage F-tests
$\ln(SPILLTECH)_{t-1}$					190.7
$\ln(SPILLSIC)_{t-1}$					38.0
Firm fixed effects	No	Yes	No	Yes	Yes
No. observations	8,579	8,579	8,387	8,579	8,579

Notes: Dependent variable is ln(R&D/Sales). Standard errors (in brackets) are robust to arbitrary heteroskedasticity and serial correlation using Newey–West corrected standard errors. All columns include a full set of year dummies and controls for current and lagged industry sales in each firm's output industry. Column (5) uses instrumental variable estimation. "1st Stage F-Tests" are the joint significance of the excluded tax-based instrumental variables (ln(*TECHTAX*) and ln(*SICTAX*)) from each first stage of the endogenous variables, ln(*SPILLTECH*) and ln(*SPILLSIC*). See Appendix B.3 for details. In column (5), we also include the firm's own federal and state R&D tax credit values.

Table VI: R&D equation

TABLE VII

COMPARISON OF EMPIRICAL RESULTS TO MODEL WITH TECHNOLOGICAL SPILLOVERS AND PRODUCT MARKET RIVALRY

(1)	(2) Partial Correlation	(3) Theory	(4) Empirics Jaffe	(5) Empirics Mahalanobis	(6) Empirics Jaffe, IV	(7) Consistency?
$\partial V_0/\partial r_\tau$	Market value with *SPILLTECH*	Positive	0.381**	0.903**	1.079***	Yes
$\partial V_0/\partial r_m$	Market value with *SPILLSIC*	Negative	−0.083**	−0.136**	−0.235**	Yes
$\partial k_0/\partial r_\tau$	Patents with *SPILLTECH*	Positive	0.417**	0.530**	0.407**	Yes
$\partial k_0/\partial r_m$	Patents with *SPILLSIC*	Zero	0.043	0.053	0.037	Yes
$\partial y_0/\partial r_\tau$	Productivity with *SPILLTECH*	Positive	0.191**	0.264**	0.206**	Yes
$\partial y_0/\partial r_m$	Productivity with *SPILLSIC*	Zero	−0.005	−0.007	0.030	Yes
$\partial r_0/\partial r_\tau$	R&D with *SPILLTECH*	Ambiguous	0.100	−0.176*	0.138	
$\partial r_0/\partial r_m$	R&D with *SPILLSIC*	Ambiguous	0.083**	0.224**	−0.022	

Notes: The theoretical predictions are for the case of technological spillovers. The empirical results are from the static fixed effects specifications for each of the dependent variables. ** denotes significance at the 5% level and * denotes significance at the 10% level (note that coefficients are as they appear in the relevant tables, not marginal

Table VII: empirical results versus theory

6.5 Tables VIII and IX: Robustness and return calculations

Read - Table VIII:

- Table VIII replaces the baseline Jaffe spillover measures with alternative measures, including Ellison-Glaeser co-agglomeration, geographic spillovers, and Jaffe covariance/exposure measures.
- The qualitative conclusions are broadly robust: technology-spillover measures remain positive in the value and productivity equations, while product-market rivalry remains negative in the value equation.
- Geographic measures show that spatial proximity can matter, but it does not overturn the core technology-versus-product-market distinction.

Read - Table IX:

- Table IX converts the empirical estimates into private and social returns to R&D.
- In the baseline Jaffe calculation, the private return is 20.7% and the social return is 55.0%, a wedge of 34.3 percentage points.
- Using the Mahalanobis measure yields an even larger wedge: 27.6% private return versus 73.7% social return.
- The wedge is larger for the largest firms because they are more connected in technology space and therefore generate larger spillovers.

Economic message: Technology spillovers dominate product-market rivalry in welfare terms. The private market under-incentivizes R&D because firms do not internalize the knowledge benefits they create for other firms.

TABLE VIII

ALTERNATIVE WAYS OF MEASURING SPILLOVERS

Dependent Variable:	(1) Tobin's Q	(2) Cite Weighted Patents	(3) Real Sales	(4) R&D/Sales
A. Baseline (Summarized From Tables III–VI Above)				
$\ln(SPILLTECH)_{t-1}$	0.381	0.468	0.191	0.100
	(0.113)	(0.080)	(0.046)	(0.076)
$\ln(SPILLSIC)_{t-1}$	−0.083	0.056	−0.005	0.083
	(0.032)	(0.037)	(0.011)	(0.034)
Observations	9,944	9,023	9,935	8,579
B. Spillovers Based on Ellison–Glaeser Co-Agglomeration Method				
$\ln(SPILLTECH^{EG})_{t-1}$	0.961	0.123	0.179	−0.082
	(0.181)	(0.562)	(0.073)	(0.109)
$\ln(SPILLSIC^{EG})_{t-1}$	−0.087	0.066	0.005	0.107
	(0.031)	(0.042)	(0.012)	(0.033)
Observations	9,944	9,023	9,935	8,579
C. Geographically Based Measure of Spillovers				
$\ln(SPILLTECH^{GEOG})_{t-1}$	1.314	0.037	0.117	
	(0.176)	(0.053)	(0.066)	
$\ln(SPILLTECH)_{t-1}$	−0.559	0.391	0.101	
	(0.163)	(0.069)	(0.060)	
$\ln(SPILLSIC^{GEOG})_{t-1}$	0.110			−0.041
	(0.078)			(0.094)
$\ln(SPILLSIC)_{t-1}$	−0.175			0.135
	(0.062)			(0.086)
Observations	9,944	9,122	10,018	8,579
D. Spillovers Based on Jaffe Covariance/Exposure Distance Metrics				
$\ln(SPILLTECH^{J\text{-}COV})_{t-1}$	0.282	0.470	0.142	0.096
	(0.102)	(0.084)	(0.041)	(0.068)
$\ln(SPILLSIC^{J\text{-}COV})_{t-1}$	−0.078	0.047	−0.006	0.084
	(0.032)	(0.026)	(0.012)	(0.035)
Observations	9,944	9,023	9,949	8,579

Notes: Panel A gives the baseline results: value equation in column (1) corresponds to Table III, column (2); patents equation in column (2) corresponds to Table IV, column (2); productivity equation in column (3) corresponds to Table V, column (2), and R&D equation in column (4) corresponds to Table VI, column (2). In Panel B, *TECH* is measured by the co-agglomeration index of Ellison and Glaeser (1997). Otherwise, all specifications are the same as in Panel A. In Panel C, the variable $SPILLTECH^{GEOG}$ uses the patenting distance weighting function between firms to scale their technology overlap. The variable $SPILLSIC^{GEOG}$ uses the sales distance function between two firms to scale their product market overlap (see Section 6.2). The equations all use the preferred specifications from the main tables (i.e., column (1) corresponds to Table III, column (2); column (2) corresponds to Table IV, column (2); column (3) corresponds to Table V, column (3), and column (4) corresponds to Table VI, column (2)). In Panel D, we use the same specifications as Panel A except we substitute the Jaffe-Covariance index for both technology ($SPILLTECH^{J\text{-}COV}$) and product market spillovers ($SPILLSIC^{J\text{-}COV}$), which is empirically identical to using the Exposure in our log-linear specification with ln(R&D) as an explanatory variable.

Table VIII: alternative spillover measures

TABLE IX

PRIVATE AND SOCIAL RETURNS TO R&D

Group of Firms	(1) Closeness Measure	(2) Private Return (%)	(3) Social Return (%)	(4) Wedge Percentage Points	(5) Median Employees	(6) Avg. SIC	(7) Avg. TECH
Closeness Measures							
1. All	Jaffe	20.7	55.0	34.3	3,000	0.015	0.038
2. All	Mahalanobis	27.6	73.7	46.1	3,000	0.030	0.174
3. All	Jaffe, IV	39.3	59.4	20.1	3,000	0.015	0.038
Size Splits							
4. Largest size quartile	Jaffe	21.1	67.1	46.0	29,700	0.015	0.054
5. Second size quartile	Jaffe	20.5	55.0	34.5	5,900	0.012	0.037
6. Third size quartile	Jaffe	20.7	50.8	30.1	1,680	0.016	0.033
7. Smallest size quartile	Jaffe	20.6	47.3	26.6	370	0.018	0.029

Notes: Numbers simulated across all firms in our sample with nonzero R&D capital stocks. We use our "preferred" systems of equations and coefficients as in Table VII. Details of calculations are in Appendix E. Columns (2) and (3) contain the private and social returns to a marginal $ of R&D and column (4) contains the absolute difference between columns (2) and (3). Column (5) reports the median number of employees in each group, and the last two columns report the average closeness measure between firms in product market space (*SIC*) and the average closeness measure in technology space (*TECH*). The first row calculates the private and social returns for the baseline estimates using exogenous R&D and the Jaffe based measures of distance (column (2), Table VII). The second row recalculates this for firms using the Mahalanobis distance measure (column (4), Table VII). The third row recalculates this using the Jaffe closeness measure with the tax credit instruments for firm-level R&D (column (5), Table VII). The next four rows recalculate these figures for firms based on their position in the employment size quartiles.

Table IX: private and social returns to R&D

6.6 Table X: Distance-measure properties

Read:

- Table X compares the Jaffe, Mahalanobis, Jaffe-covariance, exposure, and Ellison-Glaeser co-agglomeration measures.
- The comparison uses properties such as economic microfoundations, invariance to rescaling, within-field overlap, between-field overlap, non-overlapping fields, and robustness to aggregation.
- The Mahalanobis measure improves on the Jaffe measure by allowing between-field overlap, which is important when related technology fields generate knowledge flows.
- No single measure dominates all others across every desirable property.

Economic message: Measuring spillovers is not a mechanical exercise. Different measures embed different assumptions about how knowledge moves across firms and fields.

TABLE X

DESIRABLE PROPERTIES OF DISTANCE MEASURES

Name	Definition of $TECH_{ij}$	Economic Microfoundations *EMF*	Invariance to Re-Scaling *SCALE*	Within-Field Overlap *WFO*	Between-Field Overlap *BFO*	Nonoverlapping Fields *NOF*	Invariance to Aggregation Over Non-Active Fields *AGG*	Robustness to Aggregation of Active Fields *ROB*
Jaffe	$\frac{F_i' F_j}{\sqrt{F_i}\sqrt{F_j}}$		X	X			X	X
Mahalanobis	$\frac{F_i' \Omega F_j}{\sqrt{F_i}\sqrt{F_j}}$		X	X	X		X	X
Jaffe-Covariance	$F_i' F_j$		X	X		X	X	
Exposure	$F_i' F_j n_i$	X	X[a]	X		X	X	
Ellison–Glaeser Co-agglomeration	$\frac{\sum_\tau (s_{\tau i} - x_\tau)(s_{\tau j} - x_\tau)}{1 - \sum_\tau x_\tau^2}$	X	X	X				

Notes: The table compares the desirable theoretical properties of distance metrics as discussed in Section 7. Note that in constructing *SPILLTECH*, the *TECH* measure is multiplied by the R&D stock of firm j and then summed across all j. F_i denotes the vector of the shares of firm i's patenting in different technology fields, and Ω is the Mahalanobis matrix summarizing the co-location of technology fields. An "X" denotes that the distance measure has the indicated property, whereas a blank indicates that it does not.

[a] Although the Exposure index does not satisfy *SCALE* in a levels specifications, it does satisfy *SCALE* in a log specification (as in this paper).

Table X: desirable properties of distance measures

7 Short summary

- The paper studies whether outside R&D helps firms through technology spillovers or hurts them through product-market rivalry.
- It constructs two firm-level spillover pools: *SPILLTECH* based on patent-technology proximity and *SPILLSIC* based on product-market proximity.
- The theory predicts that technology spillovers should raise market value, patents, and productivity, while product-market rivalry should lower market value but not directly raise patents or productivity.
- The empirical results match this pattern: *SPILLTECH* is positive in the market value, patent, and productivity equations; *SPILLSIC* is negative in the market value equation and mostly insignificant in patent/productivity equations.
- The R&D equation gives weaker evidence that product-market rivals' R&D may be strategically complementary with own R&D.
- The results are robust to alternative distance measures and to IV specifications using R&D tax incentives.
- Social returns to R&D exceed private returns by a large margin, implying aggregate under-investment.
- Larger, more technologically connected firms generate higher estimated social returns than small niche firms in this sample.

8 Conclusion

8.1 Core conclusion

Bloom, Schankerman, and Van Reenen show that R&D spillovers are not one-dimensional. The same outside R&D stock can have very different effects depending on whether the outside firm is close in technology space or product-market space. Technology-neighbor R&D raises firm value, innovation, and productivity. Product-market-rival R&D lowers firm value through business stealing. Quantitatively, the positive technology-spillover channel dominates, so social returns to R&D are much larger than private returns.

8.2 Economic intuition

The intuition is that firms benefit from ideas produced nearby in technology space, but they are harmed when product-market rivals innovate. Patents and productivity help distinguish the two channels: if outside R&D raises patents and productivity, it looks like knowledge spillovers; if it lowers market value without raising patents or productivity, it looks like rivalry. Because the social planner values knowledge created for other firms but not the redistribution of profits across rivals, the social return exceeds the private return.

8.3 Incremental contribution revisited

The paper advances the literature in three major ways. First, it separates technology spillovers from product-market rivalry using distinct firm-level proximity measures. Second, it derives and tests a multi-outcome sign pattern rather than relying on a single production-function spillover regression. Third, it builds the estimates into welfare calculations that quantify private and social returns to R&D and compare alternative distance metrics.

8.4 Policy implications

The results support policies that encourage more R&D because private incentives are below social incentives. However, the paper also cautions that the strongest spillovers may come from technologically connected firms, not necessarily from small firms. This does not mean small firms should never receive support; liquidity constraints, entry, entrepreneurship, and appropriability may justify small-firm subsidies. But the spillover argument alone does not automatically favor small firms.

8.5 Limitations

The sample consists of publicly traded firms, so it is not ideal for studying very small private firms. The proximity measures are constructed from firms' patenting and sales profiles, which may evolve endogenously. Productivity is measured with sales deflated by industry prices, so firm-specific markups can matter. The IV strategy relies on tax-price variation that must affect outcomes only through R&D and spillovers. Finally, the R&D strategic-complementarity evidence is less robust than the value, patent, and productivity evidence.

8.6 Takeaway

The enduring takeaway is that R&D has both positive and negative external effects, but the positive knowledge-spillover effect is larger in this setting. The paper turns the broad idea of R&D spillovers into a disciplined empirical framework: locate firms in technology space, locate them in product-market space, test the predicted signs across outcomes, and use the estimates to quantify the wedge between private and social incentives.